Integers & The Coordinate Plane: Negative Numbers & Zero Pairs

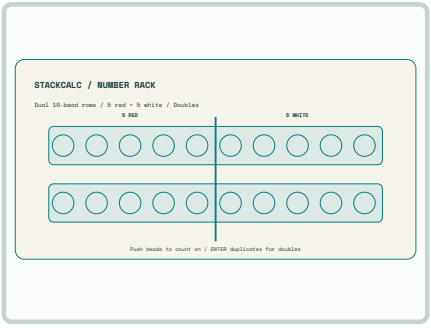
Tactile Concrete-Pictorial-Abstract (CPA) Station Guide • CCSS.MATH.6.NS.C.6

Student Name: _____ Date: _____ **Model: SC-6** **Check Score: ___ / 4**

STUDENT SELF-CHECK RULE (NO CHEATING):
Build with blocks and solve on paper first! Use your SC-6 calculator ONLY to verify answers on the 4-line LCD screen.

STATION LAB TUTORIAL: BENCH SCHEMATIC & CPA PROGRESSION

1. 2D Bench Blueprint



Kit: Number Rack

2. Desk Setup (Concrete -> Pictorial)

Physical Bench Setup:
Pair 5 positive (teal) discs with 3 negative (coral) discs. Remove 3 zero-pairs; 2 remain.

Sketch Model on Paper:
Draw positive + and negative - symbols cancelling in zero pairs.

3. Step-by-Step LCD Stack Trace (SC-6)

MODE:	123	Base-10 Arithmetic	
1.	5	X: 5	Positive 5 into X
2.	ENTER	Y: 5 | X: 5	Stack lift
3.	3	Y: 5 | X: 3	Key 3 into X
4.	+/-	Y: 5 | X: -3	Negate: X becomes -3

EXPECTED LCD X: 2

1 **Solve 8 + (-5) on the stack.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

2 **Solve -5 + (-4) on the stack.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

3 **Solve 3 - 8 on the stack.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

4 **Solve -9 - (-4) on the stack.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

Station Diagnostic & Solution Key: Chapter 6

Integers & The Coordinate Plane: Negative Numbers & Zero Pairs • Manipulative Setup, Diagnostic Rubric & Verification Flow

MANIPULATIVE HARDWARE STATION INVENTORY & 3D PRINT PREPARATION

Station Kit: Number Rack & Zero-Pair Color Discs

CAD Blueprint: number_rack

Target Device: SC-6 / SC-32 Bridge

CAD / STL Source File:

Education/scad/number_rack_parts.scad

Standard: Gridfinity 42mm Pitch / 0.6mm Inset Markings

Recommended Filaments:

White Frame with Signal Red & White Beads (PLA)

#	Math Problem Prompt	Expected Math Answer	Self-Check Keystrokes & RPN Stack
1	**Solve 8 + (-5) on the stack.** Standard: CCSS.MATH.6.NS.C.6	3	[8] [ENTER] [5] [+/-] [+]
2	**Solve -5 + (-4) on the stack.** Standard: CCSS.MATH.6.NS.C.6	-9	[5] [+/-] [ENTER] [4] [+/-] [+]
3	**Solve 3 - 8 on the stack.** Standard: CCSS.MATH.6.NS.C.6	-5	[3] [ENTER] [8] [-]
4	**Solve -9 - (-4) on the stack.** Standard: CCSS.MATH.6.NS.C.6	-5	[9] [+/-] [ENTER] [4] [+/-] [-]

TEACHER DIAGNOSTIC GUIDE: COMMON MISCONCEPTIONS & REMEDIATION

Key Misconception to Target During Station Observation:

Watch for students treating subtraction and negative signs as unrelated concepts.

How the Manipulatives & Calculator Foster Self-Correction:

Students solve with pencil and physical blocks first. When they self-check on the SC-6, any discrepancy between their written answer and the calculator's 4-line LCD immediately prompts them to re-inspect their physical bench model or regrouping, turning errors into productive struggle without teacher intervention.

Station Facilitation & Pacing Tip:

Keep 3D printed trays docked on standard Gridfinity baseplates to prevent loose parts on student desks. Encourage students to articulate their CPA steps: name the block representation (Concrete), point to their drawn tape/bond model (Pictorial), and verify register values (Abstract).

CLASSROOM SCORING & STATION MASTERY RUBRIC

Exemplary (18–20 pts):	Complete, accurate manipulative sketches; exact math answers; independently verified self-check on LCD stack.
Proficient (14–17 pts):	Correct math answers; minor sketch omissions; verified on calculator with minimal teacher prompting.
Developing (10–13 pts):	Understands problem concept but made computational slips; successfully self-corrected using calculator LCD.
Intervention (<10 pts):	Relied on calculator before solving on paper, or struggled with underlying physical manipulative models.